

I study topological and non-Hermitian phenomena in condensed matter and photonic systems.

EDUCATION

PhD Physics at McGill University in Montréal, Canada	Sep. 2023 –
Supervisors: Prof. Tami Pereg-Barnea, Prof. Kai Wang	
Visiting PhD student at California Institute of Technology in Pasadena, United States	Sep. 2026 – Dec. 2026
Supervisor: Gil Refael	
MSc Math at McGill University in Montréal, Canada	May 2022 – May 2023
Supervisor: Prof. Vojkan Jakšić	
Visiting MSc Student at CY Institute for Advanced Studies in Cergy-Pontoise, France	May 2022 – Jul. 2022
Supervisor: Prof. Armen Shirikyan	
BSc Physics and Math Joint Honours at McGill University in Montréal, Canada	Sep. 2019 – May 2022

PUBLICATIONS

Research

1. C. Fortin, K. Wang and T. Pereg-Barnea, *Finite and disordered Kitaev chains: a large deviation study*, (2026) arXiv:2606.17881 .
2. C. Fortin, K. Wang and T. Pereg-Barnea, *Unifying Anderson transitions and topological amplification in non-Hermitian chains*, Phys. Rev. B **112**, 174208 (2025). arXiv:2509.05842
3. C. Fortin, K. Wang and T. Pereg-Barnea, *Topological amplification of the bosonic Kitaev chain with nonuniform loss*, Phys. Rev. B **112**, 064208 (2025). arXiv:2412.09744
4. K.E.M. Church and C. Fortin, *Computer-assisted methods for analyzing periodic orbits in vibrating gravitational billiards*, Int. J. Bifurcation Chaos **31**, 2310021 (2021). (Selected as a **Feature Article**)

Thesis

C. Fortin, *Central Limit Theorem and Large Deviations of the Maximum Likelihood Estimator*. Master's thesis. McGill University (2023).

AWARDS AND FUNDING

COMPETITIVE AWARDS

- **FRQ-NT Doctoral Research Scholarship (B2)**: \$58,884 CAD Apr. 2026 – Sep. 2028
Provincial competition of the Fonds de recherche du Québec
- **Mitacs Globalink Research Award**: \$6,000 CAD Sep. 2026 – Dec. 2026
Mobility award to perform research at the California Institute of Technology
- **Graduate Mobility Award**: \$5,115 CAD Sep. 2026 – Dec. 2026
McGill mobility award to perform research at the California Institute of Technology
- **Chalk Rowles fellowship**: \$11,120 CAD Sep. 2025 – Aug. 2026
McGill Faculty of Science fellowship awarded to two graduate students in physics.
- **Best Poster Award** at Topological Photonics and Beyond 2025, Nankai University: ¥1,000 CNY Sep. 2025
Awarded by Nature Reviews Electrical Engineering
- **Emily Ross Crawford scholarship**: \$1,000 CAD Sep. 2020
McGill Faculty of Science scholarship awarded to undergraduate candidates of high academic merit.
- **Dean's Honour list**: Sep. 2020 – May 2021
Top 10% of students in the Faculty of Science.

RESEARCH FUNDING

- **PhD Research Stipend:** \$28,000 CAD/yr *Sep. 2023 – Mar. 2026*
Supervisor-funded stipend, reduced in 2025–2026 to offset concurrent Chalk Rowles fellowship (listed above)
- **MSc Research Stipend:** \$10,655 CAD *May. 2022 – Apr. 2023*
Supervisor-funded stipend
- **Summer Research Stipend:** \$3,995 CAD *May. 2021 – Aug. 2021*
Supervisor-funded stipend

PRESENTATIONS

CONTRIBUTED TALKS

- International Workshop on Disordered Systems 2026 in Montreal, Canada *June 2026*
- APS Global Physics Summit 2026 in Denver, United States *Mar. 2026*
- APS Global Physics Summit 2025 in Anaheim, United States *Mar. 2025*
- CONFETI 2025, INTRIQ in Bromont, Canada *Jan. 2025*
- Institute of Photonic Sciences (ICFO) in the Quantum Optics Theory group in Barcelona, Spain *Oct. 2024*

POSTER SESSIONS

- Beyond Linearity: Frontiers in Non-Hermitian Many-Body Physics 2025 in Erice, Italy *Dec. 2025*
- MaQTech Winter School 2025 in Orford, Canada *Nov. 2025*
- Fall 2025 INTRIQ meeting in Bromont, Canada *Nov. 2025*
- Emerging Quantum Phenomena (RQMP) at the Institut Quantique in Sherbrooke, Canada *Oct. 2025*
- Grande Conférence de l'Institut Courtois in Montréal, Canada *Oct. 2025*
- International Workshop on Topological Photonics and Beyond 2025 in Tianjin, China *Sep. 2025*
- ICTP Conference on Advances in Topological Condensed Matter in Trieste, Italy *Nov. 2024*
- Fall 2024 INTRIQ meeting in Bromont, Canada *Oct. 2024*
- MaQTech & PQS2D Annual Retreats in Ottawa, Canada *Jul. 2024*

RELEVANT EXPERIENCE

Teaching Assistantships (McGill University)

Preparation of tutorial sessions. Substituting lectures. Holding office hours. Marking of assignments and examinations.

List of courses:

- PHYS 319 – Introduction to Biophysics *Jan. 2026 – May 2026*
- PHYS 230 – Dynamics of Simple Systems *Sep. 2025 – Dec. 2025*
- PHYS 457 – Honours Quantum Mechanics II *Jan. 2025 – May 2025*
- PHYS 551 – Quantum Theory *Sep. 2024 – Dec. 2024*
- PHYS 457 – Honours Quantum Mechanics II *Jan. 2024 – May 2024*
- PHYS 352 – Honours Electromagnetic Waves *Sep. 2023 – Dec. 2023*
- MATH 141 – Calculus 2 *Jan. 2023 – May 2023*
- MATH 455 – Honours Analysis 4 *Jan. 2023 – May 2023*
- MATH 454 – Honours Analysis 3 *Sep. 2022 – Dec. 2022*

Teaching Assistantships (CY Institute for Advanced Studies)

Preparation of weekly lectures. Marking assignments. List of courses:

- Information-theoretic notions of entropy *May 2022 – Jul. 2022*
- Complex Analysis *May 2022 – Jul. 2022*
- Finite State Markov Chains *May 2022 – Jul. 2022*

Supervisor of research projects (McGill University)

- Guidance of an undergraduate summer research internship *May 2026 – Aug. 2026*
- Guidance of an undergraduate for the honours research project course PHYS 479. *Jan. 2026 – Apr. 2026*

SciLearn Peer Collaboration Teaching Assistant (McGill University)

- Providing guidance to undergraduate students in mathematics. *Jan. 2023 – May 2023*

Math and Physics student Helpdesk (Collège de Maisonneuve)

- Providing guidance to CÉGEP students (grade 12 & 13) in calculus and classical mechanics. *Jan. 2019 – May 2019*

VOLUNTEERING AND INVOLVEMENT

CPM student committee (McGill University)

Sep. 2025 –

- Event planner for the CPM (Centre for the Physics of Materials)

Volunteer for the Physics & TSI Outreach group (McGill University)

Sep. 2023 –

- Helping run outreach activities for the general public and prospective graduate students.
- Judge for the McGill Physics Hackathon.

RQMP selection committee for AMA scholarships

Mar. 2026

- Evaluating undergraduate internship projects

Scientific article reviewing

- American Physical Society's PRL, PRR, PRA, PRB.

UNDERGRADUATE RESEARCH EXPERIENCE

McGill University, Montréal, Canada

May 2021 – Aug. 2021

Supervisor: Prof. Vojkan Jakšić

- Quantum information theory and parameter estimation of classical Markov chains.

McGill University, Montréal, Canada

Sep. 2020 – Dec. 2020

Supervisor: Prof. Daryl Haggard

- X-ray flares analysis of the supermassive black hole Sgr. A* using the Bayesian Blocks algorithm.

McGill University, Montréal, Canada

May 2020 – Aug. 2020

Supervisor: Prof. Jean-Philippe Lessard

- Computer-assisted methods for dynamical systems.

LANGUAGES

Fluent: French, English. **Intermediate:** Spanish, German. **Beginner:** Serbo-Croatian.

Programming: Mathematica, Python, MATLAB, Bash, LaTeX.